WORK, ENERGY AND POWER

Important formulas and concepts directly from the NCERT-

❖ Work-

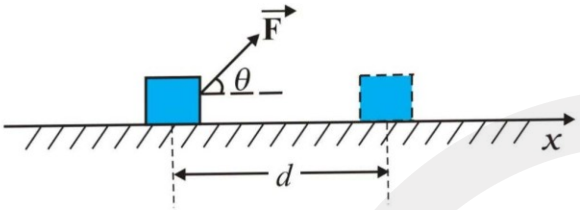

Fig. 5.2 An object undergoes a displacement **d** under the influence of the force **F**.

- Work is dot product of displacement and force.

  $$\mathbf{W = F.S = FS\cos\theta}$$

  θ = angle b/w force and displacement

- Work is given by area of force and displacement curve.

- **Work depend on frame of reference.**

- Work is a scalar quantity.

- If more than one forces are present, then net work done,

  $$\mathbf{W_{net} = F_1S_1 + F_2S_2 + F_3S_3\ldots}$$

- **Work done by centripetal force is always zero** because, centripetal force and displacement are always perpendicular to each other ($\cos 90^\circ = 0$)

- ***Work done by gravity***,

  $$\mathbf{W = mgh} \qquad \text{(h=vertical displacement)}$$

  - **This work done doesn't depend on path.**

- ***Work done by friction***,

  $$\mathbf{W = f.S = \mu N.S}$$

  - **This work done depends on path.**

- ***Work done by spring,***

$$W = \frac{1}{2} kx^2 \qquad \text{(x = elongation or compression)}$$

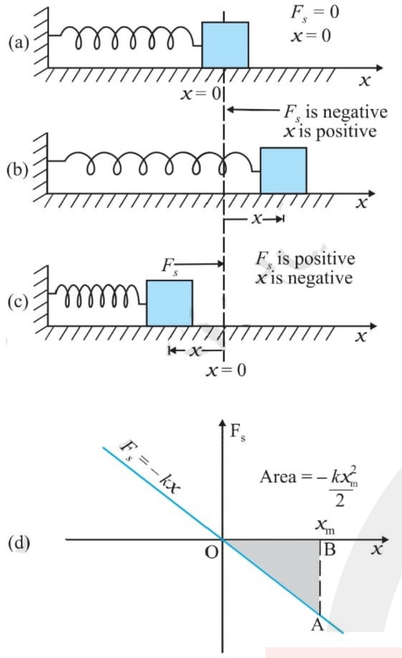

Fig. 5.7 *Illustration of the spring force with a block attached to the free end of the spring. (a) The spring force F_s is zero when the displacement x from the equilibrium position is zero. (b) For the stretched spring $x > 0$ and $F_s < 0$ (c) For the compressed spring $x < 0$ and $F_s > 0$.(d) The plot of F_s versus x. The area of the shaded triangle represents the work done by the spring force. Due to the opposing signs of F_s and x, this work done is negative, $W_s = -kx_m^2 / 2$.*

❖ <u>**Momentum-**</u>

- Motion contained in a body.
- Vector quantity.
- Direction along velocity
- $\mathbf{P = m\upsilon}$

❖ <u>**Kinetic energy-**</u>

- Energy stored due to motion.
- Scalar quantity.
- $\mathbf{K = \frac{1}{2} m\upsilon^2}$

❖ <u>**Relation b/w kinetic energy and momentum-**</u>

$$K = P^2/2m$$

$$\text{Or} \qquad P = (2mK)^{1/2}$$

- **Work energy theorem-**

- **$W_{\text{by all forces}} = \Delta K$**
- **Valid in all frame of reference.**

- **Difference b/w conservative and non-conservative force-**

CONSERVATIVE FORCE(C.F.)	NON-CONSERVATIVE FORCE(N.C.F.)
1) Its work done doesn't depend on path.	1) Its work done depends on path.
2) Potential energy stored due to negative work done by conservative force. $-W_{C.F.} = \Delta U$	2) Potential energy is not defined for non-conservative force.
3) Work done by C.F. is reversible.	3) Work done by N.C.F. is irreversible.
4) Work done in closed loop is zero.	4) Work done in closed loop may be zero.

- **Potential energy-**

- it is not a physical quantity, it doesn't have a unique value for a point, it depends on frame of reference.
- **Change in P.E. doesn't depend on frame of reference.**
- Energy stored due to shape, size and position.
- **Conservative force, $F = -dU/dr$ (potential energy gradient)**
- **If W by N.C.F. is zero then mechanical energy is constant,**

$$K + U = \text{constant}$$

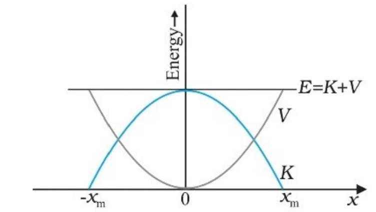

Fig. 5.8 *Parabolic plots of the potential energy V and kinetic energy K of a block attached to a spring obeying Hooke's law. The two plots are complementary, one decreasing as the other increases. The total mechanical energy E = K + V remains constant.*

- ***Gravitational potential energy,***

  U = mgh (depends on reference)

- ***Potential energy stored in spring due to elongation/compression,***

  $$U = \frac{1}{2} kx^2$$

- **<u>Vertical circular motion(VCM)-</u>**

- **Velocity at any point of VCM,**

  $$v = [v_o^2 - 2gl(1-\cos\theta)]^{1/2}$$ (v_o = velocity at lowermost point)

- **Tension at any point of VCM,**

  $$T = mv^2/l + mg\cos\theta$$

  Or $$T = mv_o^2/l - 2mg + 3mg\cos\theta$$ (vo = velocity at lowermost Point)

- **Minimum velocity(v_{min}) at lowermost point of string to complete VCM is $(5gl)^{1/2}$**

- **If $v > v_{min}$ then tension at top > 0 and velocity at topmost point > $(gl)^{1/2}$**

  If $v = v_{min}$ then tension at top = 0 and velocity at topmost point = $(gl)^{1/2}$

- **<u>Power-</u>**

- Power is rate of change of work done.

- **Average power (in time t),**

  $$P_{avg.} = W_{total}/t_{total}$$
  $$= \Delta K/t_{total}$$

- **Instantaneous power (at time t),**

  $$P_{inst.} = dW/dt$$
  = slope of work time graph

- **$P = F.v = Fv\cos\theta$**

- **If power is constant,**

$$\text{Velocity} \propto (t)^{1/2}$$

$$\text{Displacement} \propto (t)^{3/2}$$

- ***Power of pump*,**

$$P = \rho A \upsilon^3$$

- **If M = dm/dl, then *power* ,**

$$P = M\upsilon^3$$

Note- 1KWh = 3.6×10^6 J

Collisions-

- Interaction b/w two or more objects in which net external force is zero.

- **In collision, net external force is zero, thus momentum of the system is conserved.**

- Relative velocity of approach = u_1-u_2

  Relative velocity of separation = v_2-v_1

- **Coefficient of restitution, $e = v_2\text{-}v_1/u_1\text{-}u_2$**

- For elastic collision, **e = 1**
  - **Total momentum and total kinetic energy conserved.**

- For inelastic collision, **$0 < e < 1$**
  - **Total momentum conserved but total kinetic energy not conserved(KE loss).**

$$KE_{\text{before collision}} > KE_{\text{after collision}}$$

- For perfectly inelastic collision, **e = 0**
  - **Total momentum conserved but total kinetic energy not conserved(KE loss).**

  - **Particles move with common velocity after collision,**

$$v_{\text{common}} = m_1u_1 + m_2u_2 + \ldots / m_1 + m_2 + \ldots$$

- **Change in KE (loss),**

$$\Delta KE = \frac{1}{2}\left[m_1 m_2 (u_{rel})^2/(m_1+m_2)\right](1-e^2)$$

- **Physical contact is not necessary for collisions.**

- **Total mechanical energy is conserved in all types of collisions.**

- In case of elastic collision b/w two objects in which mass of one object is very much greater than other($m_1 >>> m_2$), then, **velocity of heavier object remains always unchanged .**

- **Fraction of KE transferred in elastic collision in which one particle is at rest and one in motion before collision,**

$$KE_{transferred}/KE_{initial} = 4n/(1+n)^2$$ (n = mass of havier particle/mass of lighter particle)

- **Fraction of KE retained in one particle which is in motion in elastic collision (other particle at rest),**

$$KE_{retained}/KE_{initial} = (n-1)^2/(n+1)^2$$

- **Elastic collision of two objects having same mass, velocities exchange after collision.**

- **In ball-ground collision,**

    - **Velocity , $v_{after\ collision(A/C)} = e^N v_{before\ collision(B/C)}$ ($e = v_{A/C}/v_{B/C}$) (N = no. of collisions)**

    - **Time, $t_{A/C} = e^N t_{B/C}$**

    - **Height attained by ball, $h_{A/C} = e^{2N} h_{B/C}$**

    - **Energy , $E_{A/C} = E_{B/C}$**

SUMMARY

1. The *work-energy theorem* states that the change in kinetic energy of a body is the work done by the net force on the body.

$$K_f - K_i = W_{net}$$

2. A force is *conservative* if (i) work done by it on an object is path independent and depends only on the end points $\{x_i, x_j\}$, or (ii) the work done by the force is zero for an arbitrary closed path taken by the object such that it returns to its initial position.
3. For a conservative force in one dimension, we may define a *potential energy* function $V(x)$ such that

$$F(x) = -\frac{\mathrm{d}V(x)}{\mathrm{d}x}$$

or

$$V_i - V_f = \int_{x_i}^{x_f} F(x)\,\mathrm{d}x$$

4. The principle of conservation of mechanical energy states that the total mechanical energy of a body remains constant if the only forces that act on the body are conservative.
5. The *gravitational potential energy* of a particle of mass m at a height x about the earth's surface is

$$V(x) = m\,g\,x$$

where the variation of g with height is ignored.

5. The elastic potential energy of a spring of force constant k and extension x is

$$V(x) = \frac{1}{2}k\,x^2$$

7. The scalar or dot product of two vectors **A** and **B** is written as **A.B** and is a scalar quantity given by :**A·B** = $AB\cos\theta$, where θ is the angle between **A** and **B**. It can be positive, negative or zero depending upon the value of θ. The scalar product of two vectors can be interpreted as the product of magnitude of one vector and component of the other vector along the first vector. For unit vectors :

$$\hat{\mathbf{i}}\cdot\hat{\mathbf{i}} = \hat{\mathbf{j}}\cdot\hat{\mathbf{j}} = \hat{\mathbf{k}}\cdot\hat{\mathbf{k}} = 1 \text{ and } \hat{\mathbf{i}}\cdot\hat{\mathbf{j}} = \hat{\mathbf{j}}\cdot\hat{\mathbf{k}} = \hat{\mathbf{k}}\cdot\hat{\mathbf{i}} = 0$$

Scalar products obey the commutative and the distributive laws.

POINTS TO PONDER

1. The phrase 'calculate the work done' is incomplete. We should refer (or imply clearly by context) to the work done by a specific force or a group of forces on a given body over a certain displacement.
2. Work done is a scalar quantity. It can be positive or negative unlike mass and kinetic energy which are positive scalar quantities. The work done by the friction or viscous force on a moving body is negative.
3. For two bodies, the sum of the mutual forces exerted between them is zero from Newton's Third Law,
$$\mathbf{F}_{12} + \mathbf{F}_{21} = 0$$
But the sum of the work done by the two forces need not *always cancel, i.e.*
$$W_{12} + W_{21} \neq 0$$
However, it *may sometimes be true.*
4. The work done by a force can be calculated sometimes even if the exact nature of the force is not known. This is clear from Example 5.2 where the WE theorem is used in such a situation.
5. The WE theorem is not independent of Newton's Second Law. The WE theorem may be viewed as a scalar form of the Second Law. The principle of conservation of mechanical energy may be viewed as a consequence of the WE theorem for conservative forces.
5. The WE theorem holds in all inertial frames. It can also be extended to non-inertial frames provided we include the pseudoforces in the calculation of the net force acting on the body under consideration.
7. The potential energy of a body subjected to a conservative force is always undetermined upto a constant. For example, the point where the potential energy is zero is a matter of choice. For the gravitational potential energy mgh, the zero of the potential energy is chosen to be the ground. For the spring potential energy $kx^2/2$, the zero of the potential energy is the equilibrium position of the oscillating mass.
8. Every force encountered in mechanics does not have an associated potential energy. For example, work done by friction over a closed path is not zero and no potential energy can be associated with friction.
9. During a collision : (a) the total linear momentum is conserved at each instant of the collision ; (b) the kinetic energy conservation (even if the collision is elastic) applies after the collision is over and does not hold at every instant of the collision. In fact the two colliding objects are deformed and may be momentarily at rest with respect to each other.